Kernix 0.10.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

31 workloads

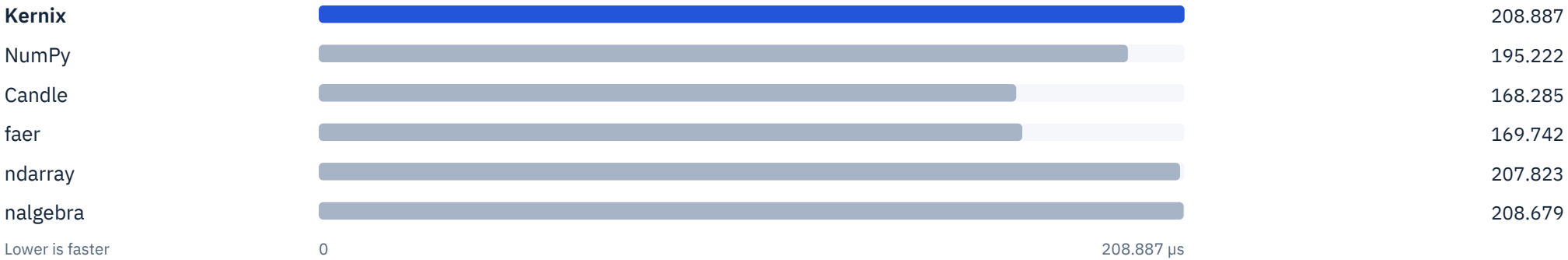
108 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

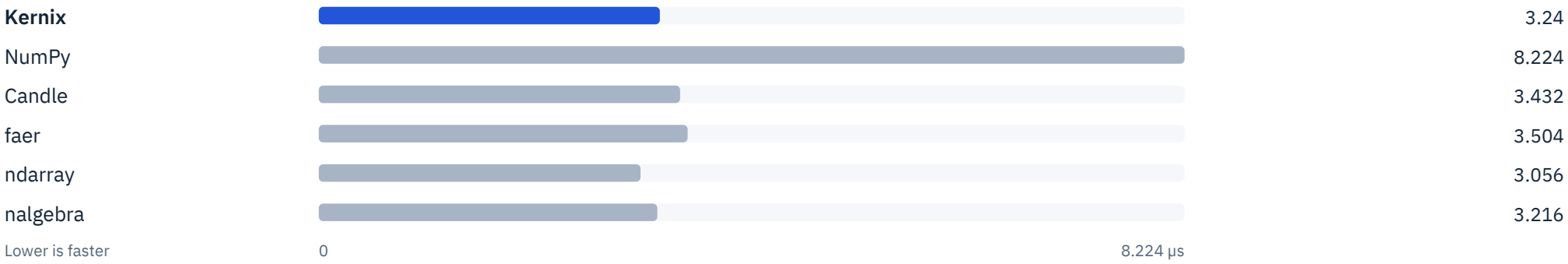
Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

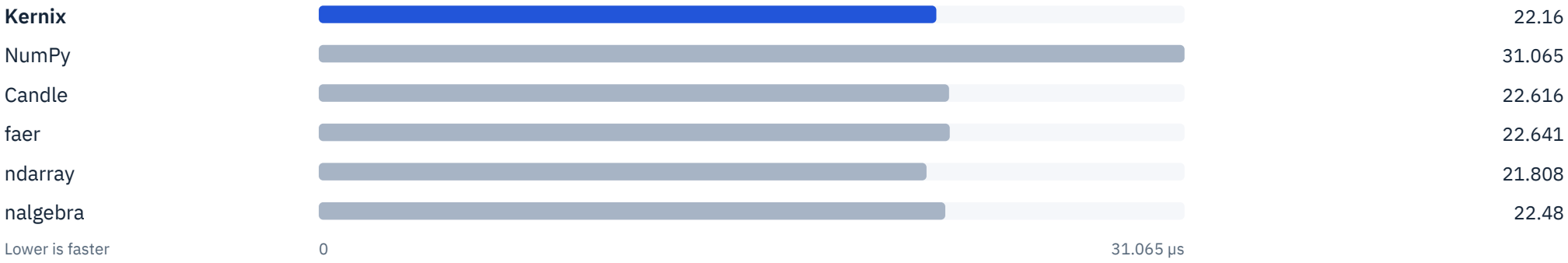
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



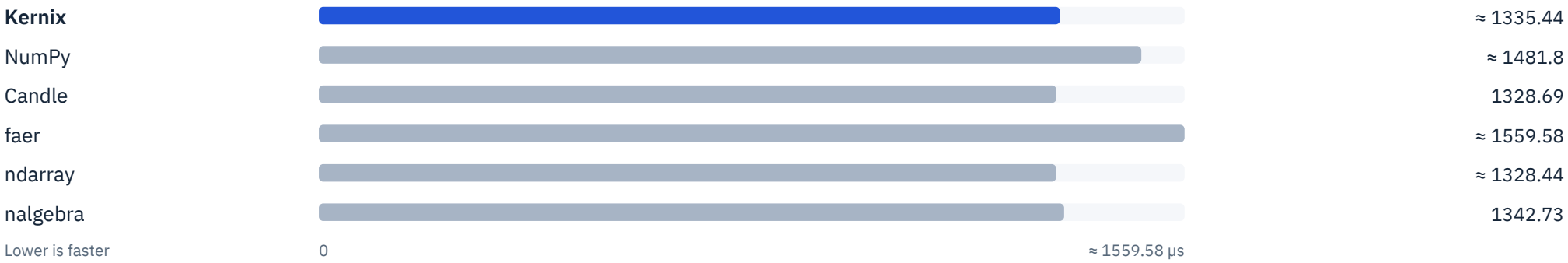
Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

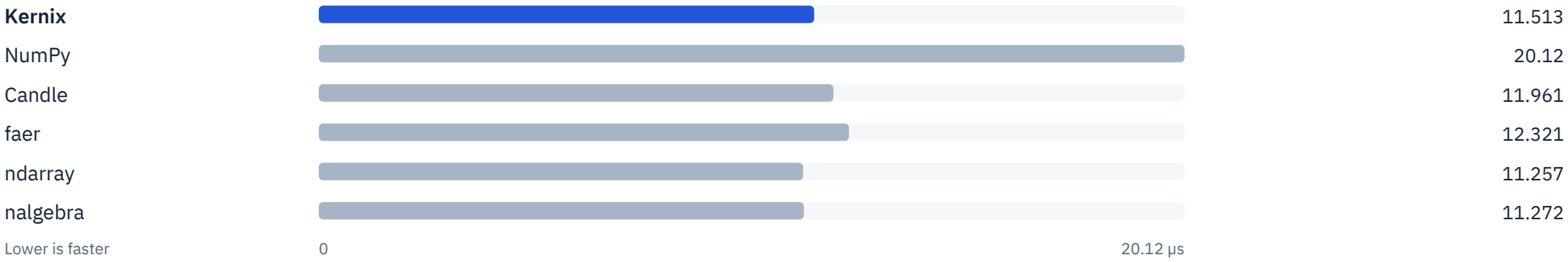
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



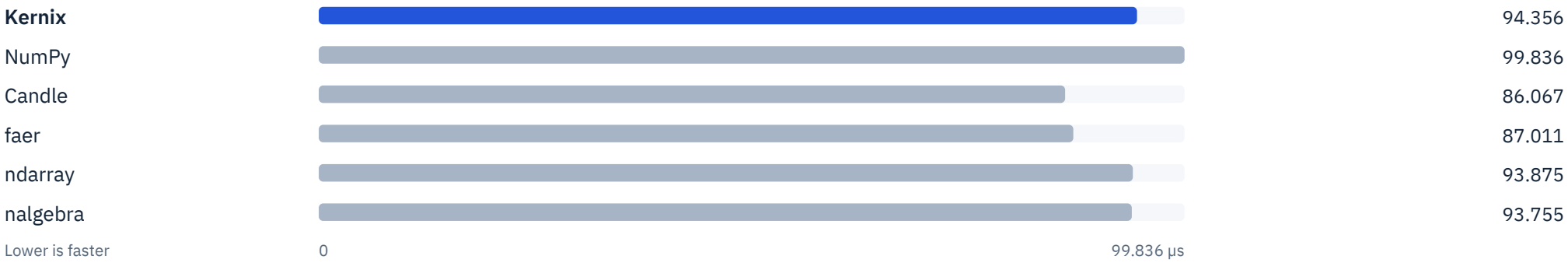
Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

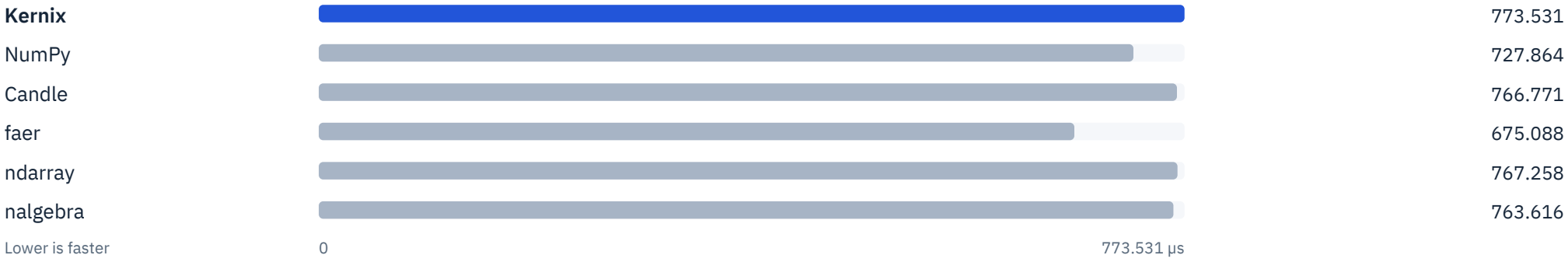
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



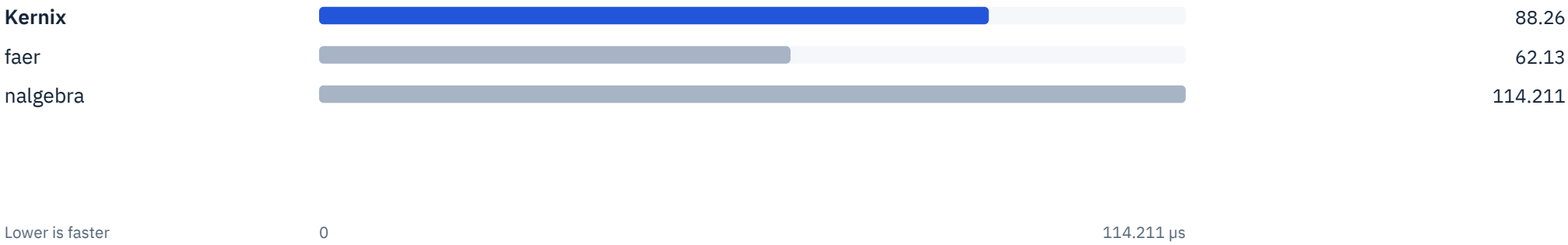
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

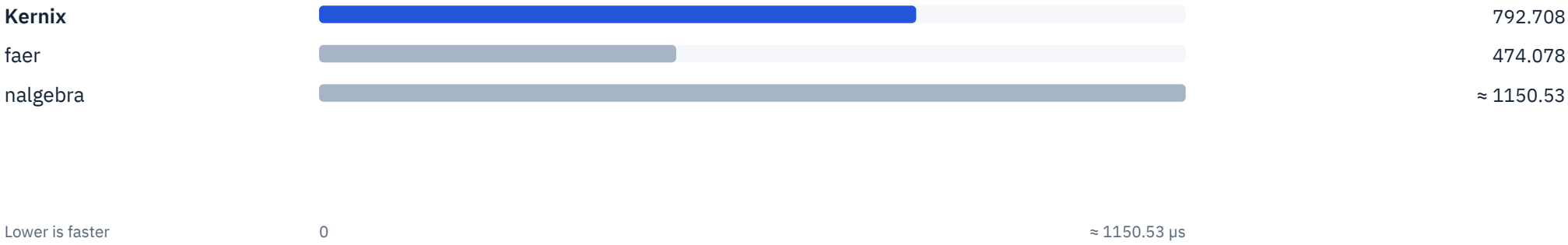
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



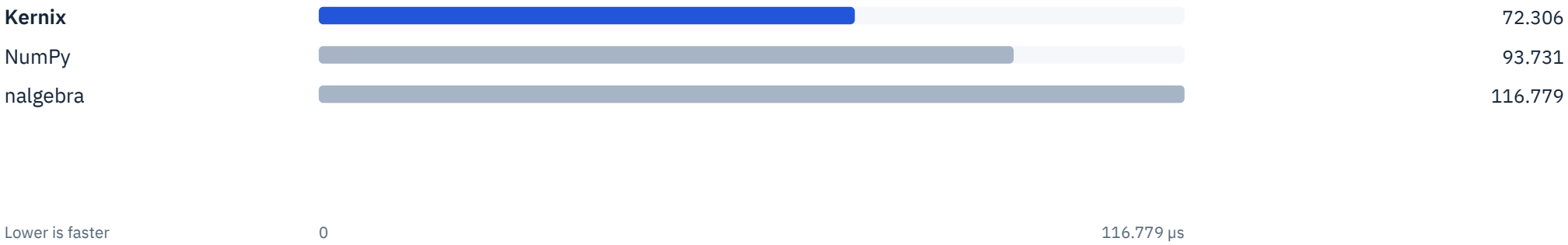
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

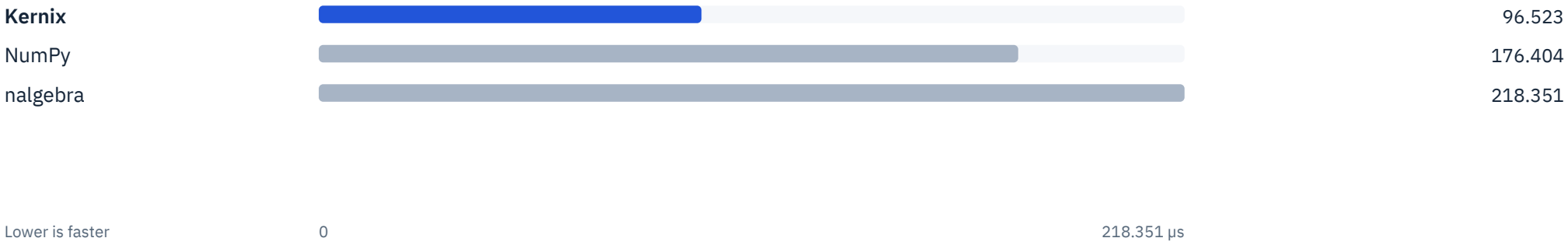
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



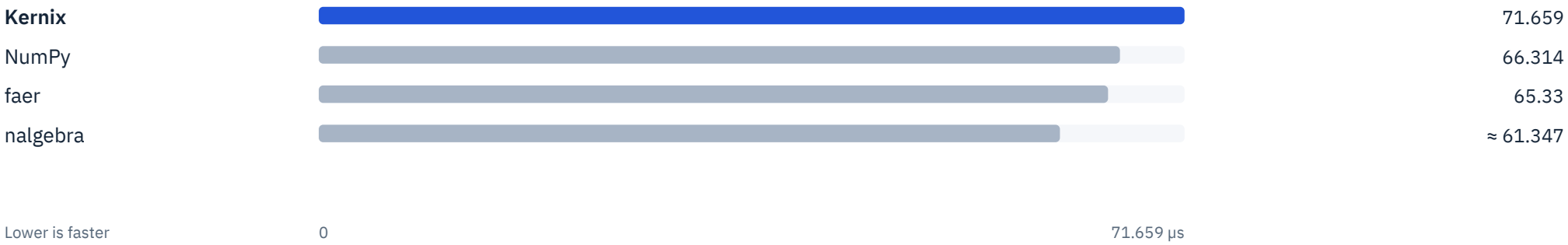
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

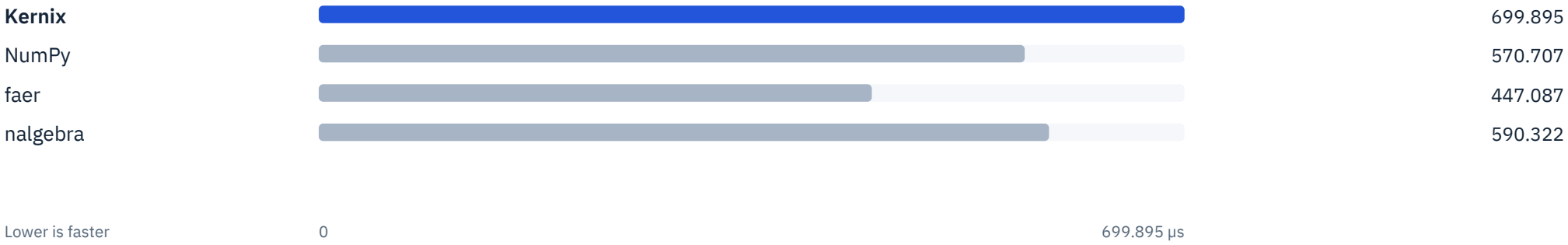
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



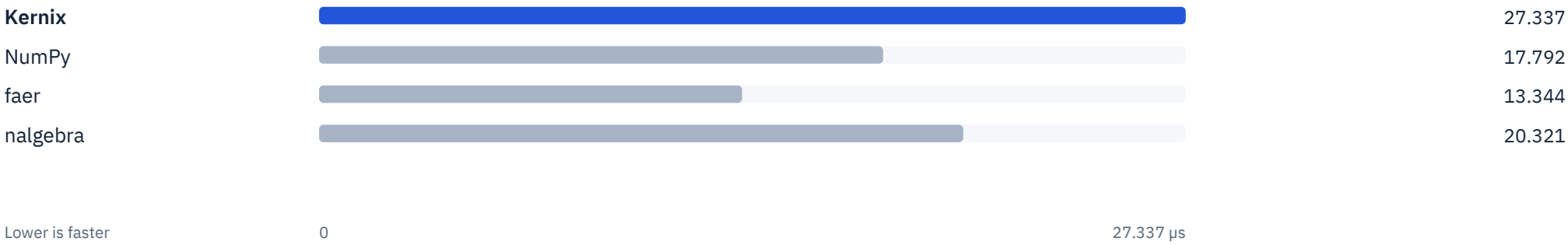
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

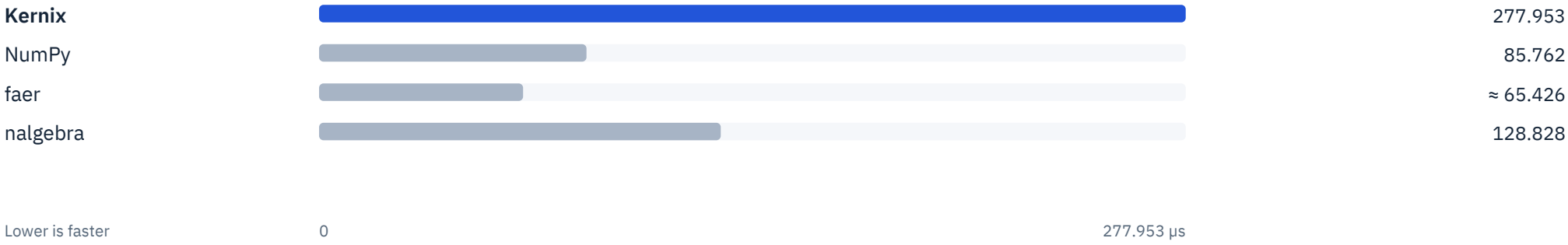
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



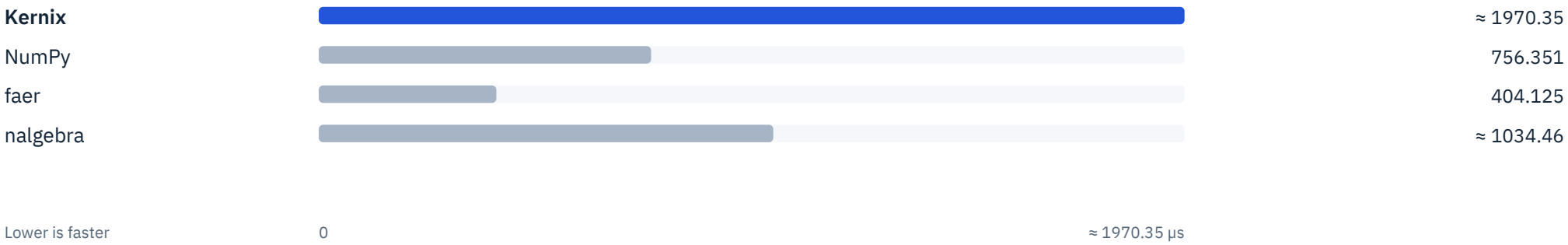
Kernix 0.10.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



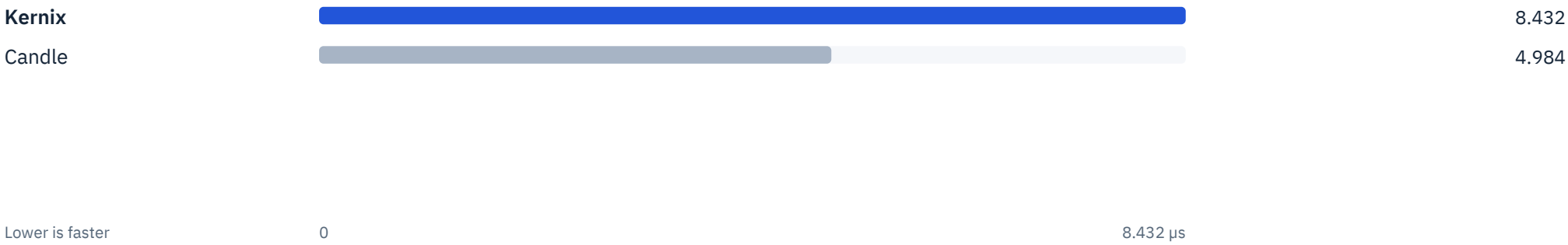
Kernix 0.10.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

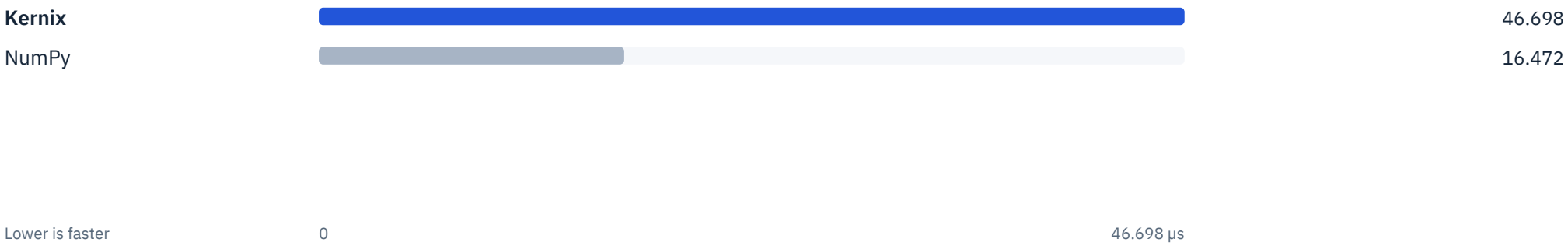


Kernix 0.10.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

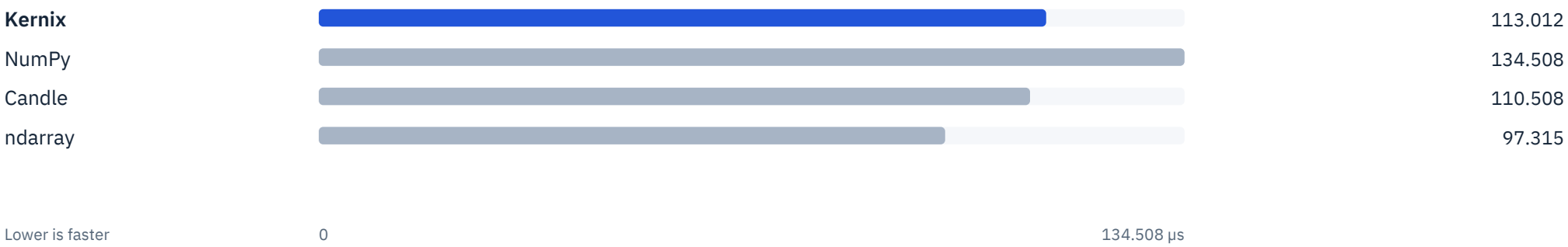


Kernix 0.10.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



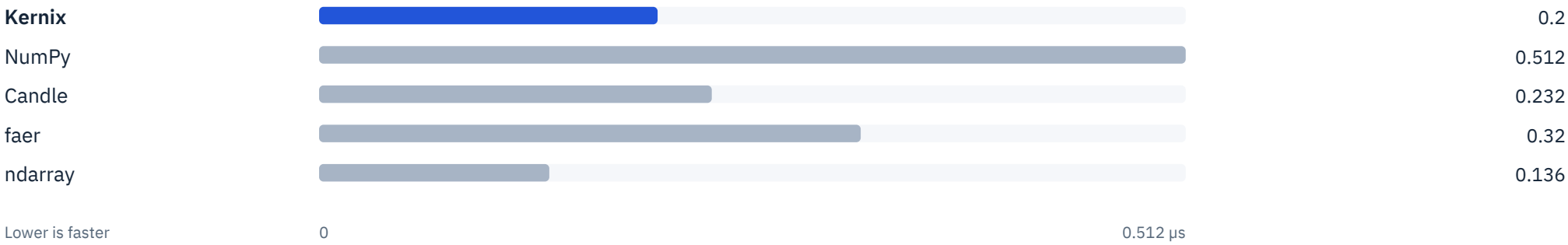
Kernix 0.10.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3504	3.504
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3432	3.432
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8224	8.224
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3240	3.24
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3056	3.056
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3216	3.216
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21808	21.808
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22641	22.641
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22616	22.616
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22160	22.16

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31065	31.065
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22480	22.48
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	208887	208.887
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169742	169.742
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	207823	207.823
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	208679	208.679
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168285	168.285
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	195222	195.222
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1335439	1335.439
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1481796	1481.796

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1328690	1328.69
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1559577	1559.577
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1342730	1342.73
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1328439	1328.439
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11272	11.272
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11961	11.961
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11513	11.513
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12321	12.321
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11257	11.257
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20120	20.12

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87011	87.011
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99836	99.836
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	86067	86.067
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93875	93.875
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	94356	94.356
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93755	93.755
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	675088	675.088
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	773531	773.531
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	766771	766.771
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	763616	763.616

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	767258	767.258
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	727864	727.864
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4441	4.441
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12984	12.984
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3839.9999999999995	3.8399999999999995
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16832	16.832
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	97835	97.835
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	841275	841.275
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	49722	49.722
Column-pivoted QR / Kernix 96 × 32 · f64	88260	88.26

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	62130	62.13
Column-pivoted QR / nalgebra 96 × 32 · f64	114211	114.211
Column-pivoted QR / Kernix 256 × 64 · f64	792708	792.708
Column-pivoted QR / faer 256 × 64 · f64	474078	474.078
Column-pivoted QR / nalgebra 256 × 64 · f64	1150531	1150.531
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72306	72.306
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	116779	116.779
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93731	93.731
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218351	218.351
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96523	96.523

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176404	176.404
Thin QR / nalgebra 96 × 32 · f64	61346.99999999999	61.34699999999999
Thin QR / NumPy 96 × 32 · f64	66314	66.314
Thin QR / Kernix 96 × 32 · f64	71659	71.659
Thin QR / faer 96 × 32 · f64	65330	65.33
Thin QR / faer 256 × 64 · f64	447087	447.087
Thin QR / Kernix 256 × 64 · f64	699895	699.895
Thin QR / NumPy 256 × 64 · f64	570707	570.707
Thin QR / nalgebra 256 × 64 · f64	590322	590.322
Cholesky factorization / NumPy 64 × 64 · f64	17792	17.792

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	27337	27.337
Cholesky factorization / faer 64 × 64 · f64	13344	13.344
Cholesky factorization / nalgebra 64 × 64 · f64	20321	20.321
Cholesky factorization / nalgebra 128 × 128 · f64	128828	128.828
Cholesky factorization / NumPy 128 × 128 · f64	85762	85.762
Cholesky factorization / faer 128 × 128 · f64	65425.99999999999	65.42599999999999
Cholesky factorization / Kernix 128 × 128 · f64	277953	277.953
Cholesky factorization / nalgebra 256 × 256 · f64	1034459	1034.459
Cholesky factorization / faer 256 × 256 · f64	404125	404.125
Cholesky factorization / Kernix 256 × 256 · f64	1970346	1970.346

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	756351	756.351
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4984	4.984
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8432	8.432
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7232	7.232
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3496	3.496
Model inference / Candle batched normalized/sliced MLP · f64	5296	5.296
Model inference / Kernix batched normalized/sliced MLP · f64	3024	3.024
Model recording + forward / Candle batched normalized/sliced MLP · f64	7664	7.664
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3817	3.817
Model backward / Kernix prepared graph → fresh gradients · f64	4200	4.2

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Candle prepared graph → fresh gradients · f64	21400	21.4
Parameter update / Kernix transactional SGD, fixed gradients · f64	720	0.72
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46698	46.698
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16472	16.472
Row softmax / Candle 256 × 64 · f64	110508	110.508
Row softmax / ndarray 256 × 64 · f64	97315	97.315
Row softmax / Kernix 256 × 64 · f64	113012	113.012
Row softmax / NumPy 256 × 64 · f64	134508	134.508
Create zeros / Candle 64 × 64 · f32	232	0.232
Create zeros / faer 64 × 64 · f32	320	0.32

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	200	0.2
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / NumPy 64 × 64 · f32	512	0.512
Create ones / Kernix 64 × 64 · f32	248	0.248
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / NumPy 64 × 64 · f32	1536	1.536
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997